# Introduction to Photogrammetry

**Lecture 10**

**Digital Camera: Geometry, Calibration, and Application**

Usman Mazhar, Ph.D.

usmanmazhargeo@outlook.com

# In Today’s Class

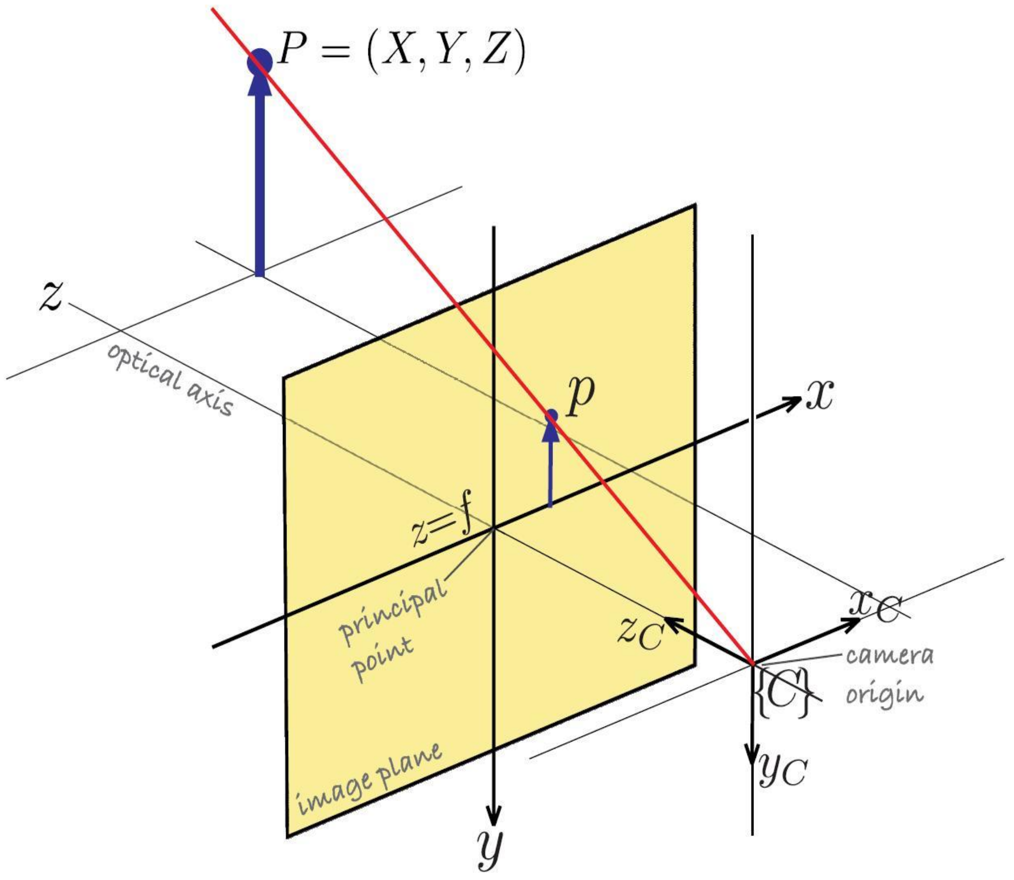
1. Camera Geometry
2. Camera Calibration
3. Applications and Advanced Concepts

# 1. Camera Geometry

- Camera geometry defines the spatial relationship between the object, camera, and image.
- It is essential for accurate measurements and 3D reconstruction.

# Interior Orientation Parameters

- Interior orientation parameters include focal length, principal point, and lens distortion. These parameters define the internal characteristics of the camera.

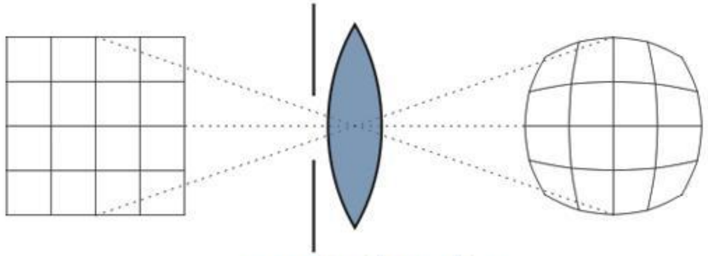


**Exterior Orientation Parameters**

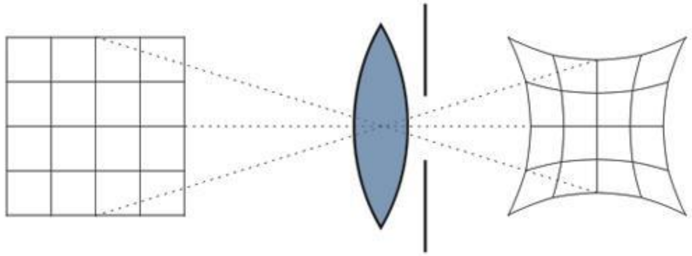
- Exterior orientation parameters describe the position and orientation of the camera in space. These include X, Y, Z coordinates and rotation angles.

## Lens Distortion

- Lens distortion causes deviations from the ideal image geometry. It includes radial and tangential distortions, which must be corrected for accurate photogrammetry



Barrel Distortion



Pincushion Distortion

# 2. Camera Calibration

- Camera calibration is the process of determining and correcting camera parameters.
- It ensures that measurements derived from images are accurate and reliable.

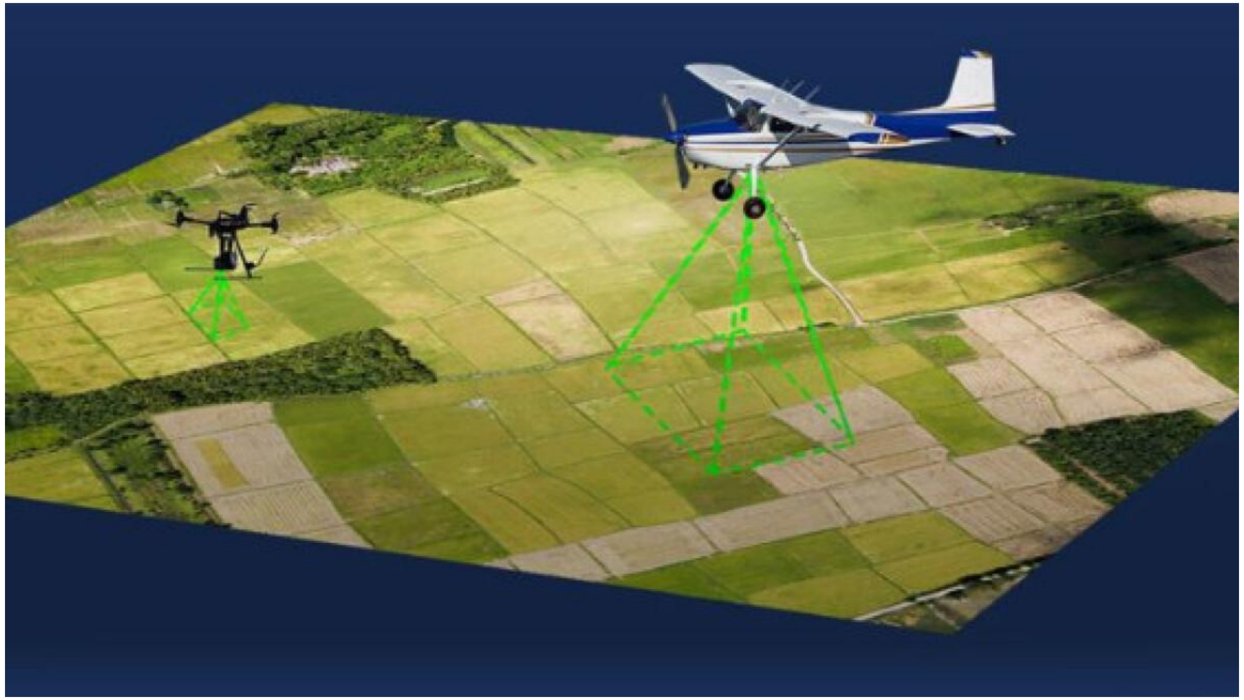
# 3. Applications and Advanced Concepts

**Aerial Photogrammetry**

- Digital cameras mounted on aircraft or drones are used to capture aerial images for mapping and terrain modeling.

## UAV and Drone Imaging

- Unmanned Aerial Vehicles (UAVs) equipped with digital cameras provide high-resolution imagery for small-area mapping and monitoring

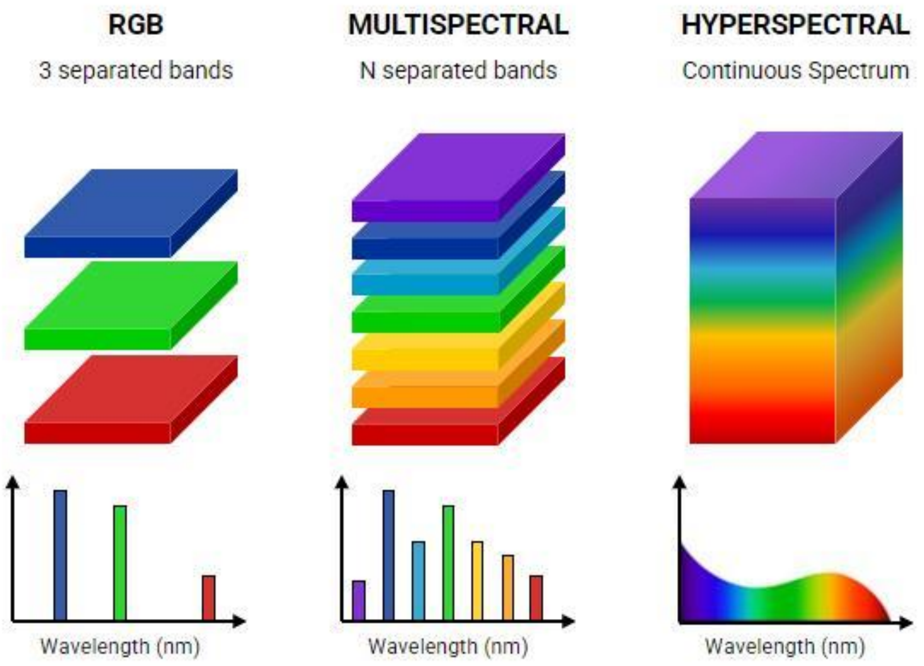


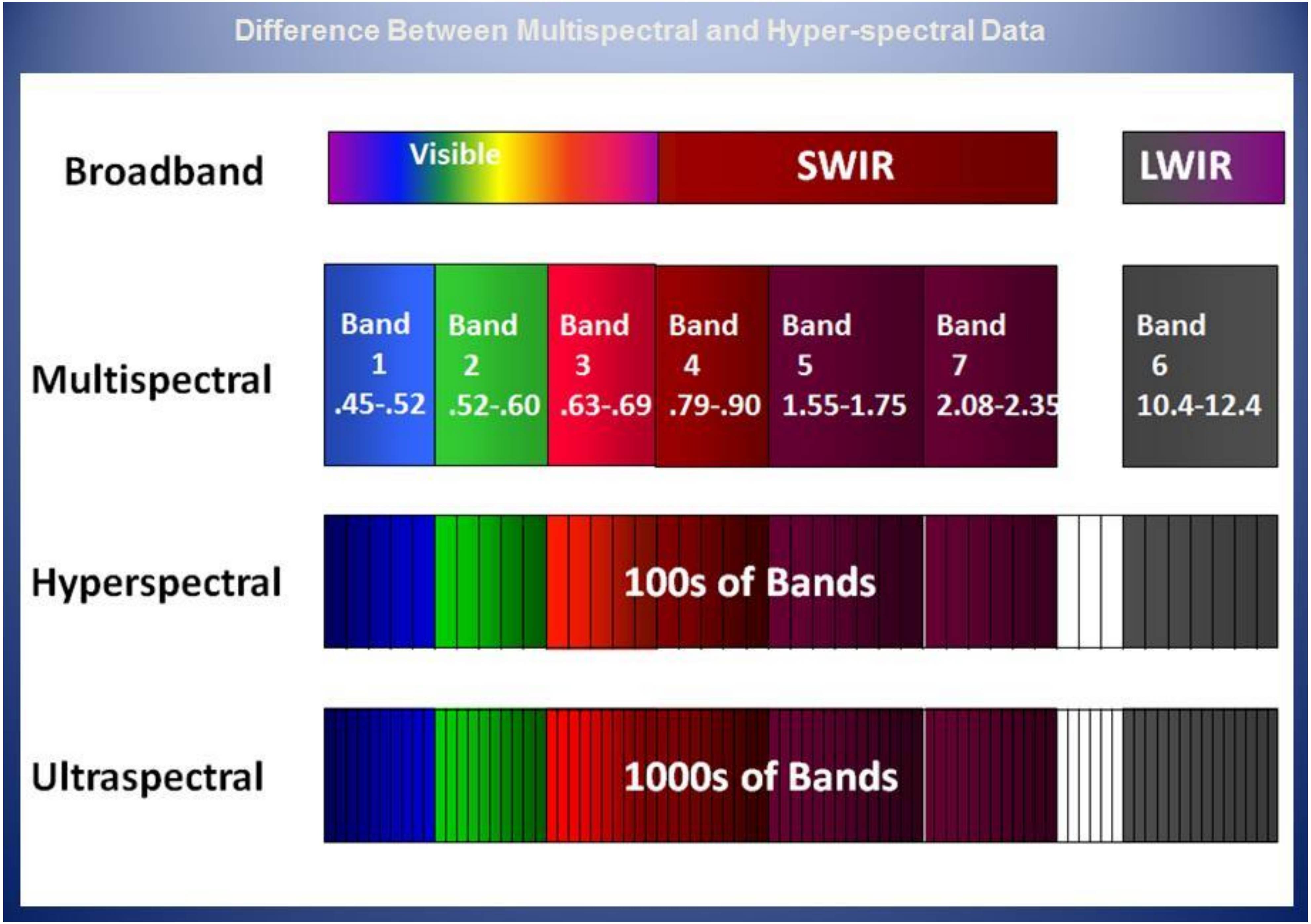
**Close-Range Photogrammetry**

- Digital cameras are used for close-range applications such as architecture, archaeology, and engineering measurements.

## Multispectral and Hyperspectral Cameras

- Advanced cameras capture data in multiple spectral bands, enabling analysis of vegetation, soil, and environmental conditions.




Difference Between Multispectral and Hyper-spectral Data
Broadband
Visible
SWIR
LWIR
Multispectral
Band 1 .45-.52
Band 2 .52-.60
Band 3 .63-.69
Band 4 .79-.90
Band 5 1.55-1.75
Band 7 2.08-2.35
Band 6 10.4-12.4
Hyperspectral
100s of Bands
Ultraspectral
1000s of Bands

## Image Quality Factors

- Image quality is influenced by factors such as lighting conditions, sensor quality, lens performance, and motion blur

- Key factors for high-quality photogrammetry include:
- **Sharpness and Focus**: Images must be sharp, with adequate depth of field to keep the subject in focus. Blurring leads to matching errors.
- **Resolution** (Megapixels): Higher resolution generally allows for better, more detailed matching, though it is not the sole indicator of quality.
- **Lighting and Contrast**: Consistent, even lighting is required to avoid deep shadows or blown-out highlights, which cause texture-less areas.

- **Image Noise**: Low sensor noise is crucial. High ISO (sensitivity) settings should be avoided to prevent grain that interferes with feature point detection.
- **Image Integrity**: RAW formats or low-compression JPGs (compression ratio < 10) are preferred to maintain necessary image data.
- **Color and Texture**: Good image contrast and texture are required for software to find corresponding points

## Future Trends

- Emerging technologies include AI-based image processing, real-time 3D reconstruction, and integration with GIS and remote sensing systems

# Thank you!

Any Questions